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PROFILE

Staff engineer with 17 years of experience across big tech, cloud infrastructure and early-stage startups. I specialise in building end-to-end features across backend and frontend, with cross-functional technical leadership that sits between engineering, product and research. I have shipped production systems operating at million RPS scale, launched a public AWS service from scratch, and spent the last four years at Meta leading and growing teams, with a recent focus on internal AI tooling.

I am drawn to hard problems where the engineering and product thinking are inseparable and I care strongly about investing in tools, automation and operational excellence.

SKILLS

BACKEND	FRONTEND	DATA & ML	LANGUAGES
Python, Java/Kotlin, Node, Hack	JS/HTML/CSS, React, Android	PyTorch, SQL, Hive, Hadoop	English, German

EMPLOYMENT

Meta — Reality Labs & Facebook

Staff Engineer / Tech Lead

July 2021 – Present · London

Founding member of the London Reality Labs Avatar org. Spent 3 years as tech lead on the Mobile Avatar Editor across Facebook, Instagram and Messenger, then moved to the Mobile Data Frameworks team to help with backend efficiency and AI-agent infrastructure work.

- Joined as a founding member of the London Reality Labs team, helping to grow the London Avatar engineering organisation from a handful of engineers to **30+ engineers**
- Supported multiple redesigns of the avatar editor across iOS and Android, driving improvements in avatar expressiveness, funnel conversion, mobile performance and overall usage rates across major surfaces (IG, FB, Messenger)
- Led the frontend work on the launch of a **ML-based avatar creation flow** using the device camera — significantly reducing time-to-creation and significantly increasing user affinity with their own avatars
- Helped set technical direction for two org-wide avatar style evolutions resulting in **two pending patents**
- Transferred to the **Mobile Data Frameworks team** in early 2025, supporting *Elastic GraphQL*: a system routing non-critical requests to opportunistic compute tiers during peak load, delivering **multi-million dollar annual savings** and unlocking prefetch and refresh patterns that had previously been cost-prohibitive — driving measurable engagement improvements for surfaces like NewsFeed
- AI** In early 2026, joined an internal initiative to build an **AI-powered prototyping portal** for cross-platform mobile UI using Meta's internal cross-platform mobile framework — demonstrating that internal tooling could match the speed and quality of open-source based prototyping workflows.

Focused on the quality and verification layer: building agentic tools for autonomous screen rendering, testing and verification, and evolving the architecture from a single-agent one-shot approach to a **plan-code-verify orchestrator** with coordinated agent teams

AI Currently building Meta's first **agentic engineering team** — an experimental system blending persistent team-specific AI lead agents with ephemeral specialist agents for individual tasks, covering maintenance, operations and feature development

Amazon / AWS Machine Learning

Oct 2017 – May 2021 · London

Software Development Engineer

Four years across two distinct phases: Amazon's retail platform at petabyte scale, followed by the small team that built and publicly launched Amazon CodeGuru Profiler.

First 18 months in Amazon's retail inventory platform, working on services backing first- and third-party seller catalogues. The team managed one of the **three largest DynamoDB tables globally**, with engineering priorities centred on fault tolerance, operational resilience and backend performance. This is where I developed specialist expertise in JVM performance analysis and optimisation

That expertise led to joining a nascent internal team building **Amazon's profiling tooling**, allowing engineering teams across Amazon to identify and fix performance regressions. Led the frontend engineering effort, designing and building interactive visualisation renderers including interactive flame graphs, icicle graphs and time series.

As the tool gained internal traction, the team pivoted to launching it as a **public AWS service: Amazon CodeGuru Profiler**. Scope broadened from pure performance analysis to operational support — pairing profiling data with a dedicated ML team to build proactive anomaly detection. Led prototyping and validation of anomaly detection approaches through to GA launch of the service

Gamevy

Aug 2013 – Sep 2017 · London

Software Engineer / Project Manager

Day-one engineer at a London gaming startup, working across game design, frontend, backend and operations.

Developed game concepts and led full-stack delivery for the majority of Gamevy's catalogue, acting as project manager across multiple simultaneous releases

Built probabilistic models to verify statistical behaviour of game mechanics

Last.fm

Oct 2011 – Aug 2013 · London

Data Engineer

Joined as a frontend engineer and later moved to the data team.

Supported various product launches, and implemented core parts of Last.fm's real-time music charts pipeline.

EDUCATION

Baden-Württemberg Cooperative State University Mannheim

2008 – 2011

BSc Computer Science · Dual programme with IBM

INTERESTS

Volunteered for multiple years at [Electromagnetic Field](#), a UK technology and arts festival for the hackers and makers. I like fine dining, travel, and endurance running.